Image classification using Vision Transformers on small datasets

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Abstract

Vision transformers make a revolution in the field of computer vision and image classification tasks. They were characterized by their outstanding performance, which surpasses state-of-the-art convolutional neural networks. However, ViTs are criticized for their hunger for data and the need for large computational resources. While the original Vision Transformer model demonstrated impressive performance when pre-trained on JFT-300M, its effectiveness on smaller datasets without extensive computational resources remained questionable. This paper investigates the application of Vision Transformers in image classification tasks on smaller datasets without relying on large-scale pre-training. This limitation is tackled by implementing a modified Vision Transformer architecture that can achieve competitive performance on standard benchmark datasets with minimal resources. The implementation mostly follows the principles laid out by the paper "An Image is Worth 16x16 Words." However, specific adjustments to hyperparameters and other architectural components were made to suit smaller datasets. Our model uses 4x4 patches instead of the original 16×16 ones and fewer transformer encoder layers, achieving 99.5% for MNIST and 81.5% for CIFAR-10 scratch training for 60 epochs. Therefore, with appropriate modifications, Vision Transformers may well be applied to smaller datasets without the hindrance of large-scale pre-training and thus offer, in many situations and under limited computational resources, a viable alternative to CNNs.

Code – https://anonymous.4open.science/r/VisionTransformer-1D10

Introduction

Image classification is a well-known topic in computer vision, where the goal is to assign predefined categories or labels to images based on their content. There are a number of applications in almost every field, such as autonomous driving systems, healthcare diagnostics, security surveillance, industrial quality control, and content moderation on social media platforms. CNNs have traditionally been the default approach to image classification due to their ability to model spatial hierarchies and local patterns with architectures like AlexNet, ResNet, and EfficientNet. Yet CNNs are poor at capturing long-range dependencies in images, which has led researchers to explore alternative architectures. Dosovitskiy et al. introduce the Vision Transformer (ViT). The paper shows that transformers could be a good replacement for convolutional neural networks (CNNs) for image classification tasks. This represents a revolutionary shift in this field due to transformers efficiency and scalability, which makes it possible to train models with over 100 billion parameters. The image is divided into patches, and then these patches are treated like tokens. Patches are linearly projected into embeddings, and a CLS token is prepended to the sequence for classification tasks. The challenge in applying self-attention to images is that each pixel should attend to every other pixel, which would result in quadratic computational complexity. For example, a 100 × 100 image (10,000 total pixels) would require calculating self-attention 100 million times. So, to solve this issue, self-attention should be applied only in local neighborhoods for each pixel. Another problem is that transformers don't have any sense of the position of the pixels. The solution is to add a position encoding vector to the patch embedding. The patch embeddings, along with the positional encodings, are then processed by a standard transformer encoder, which consists of multi-head self-attention mechanisms and MLP (multilayer perceptron) blocks. The proposed vision transformer demonstrates excellent performance across multiple benchmark datasets. When pre-trained on the large-scale JFT-300M dataset and fine-tuned on ImageNet, the ViT-Huge model achieves an accuracy of 88.55% compared to CNN-based ResNet152x4, which achieves 87.54%. For CIFAR-100, ViT-Huge reaches an accuracy of 94.55% while CNN-based ResNet152x4 reaches 93.51%. Both the ViT-Huge and ViT-Large outperform ResNet-based when pre-trained on the JFT-300M dataset and fine-tuned on all the datasets. The primary strength of Vision Transformer lies in its scalability. Using a pure transformer architecture for vision, it surpasses the state-of-the-art CNNs performance. Also, the computational efficiency of vision transformers is impressive. The pre-trained model requires fewer computational resources in training than its CNN counterparts while giving better performance. JFT-300M Pretrained ViT-Huge requires 2.5k TPUv3-core-days, and JFT-300M pretrained ViT-Large requires 0.68k, while CNN-based ResNet152x4 requires 9.9k TPUv3-core-days. Additionally, ViT's self-attention mechanism can capture global dependencies that CNNs struggle to model efficiently.

Related Work

The Rise of Transformers in NLP

NLP tasks were dominated by recurrent neural networks (RNNs) and convolutional neural networks (CNNs). In the paper "Attention Is All You Need," Vaswani et al. introduced the transformer architecture, which overcomes their limitations in parallelization and modeling long-range dependencies efficiently. Multi-head attention is proposed in the paper, which essentially has multiple attention heads performing their multiple aspects of linguistic relationships (e.g., syntax, coreference). Since attention doesn't preserve ordering, the transformer employs positional encodings so that information regarding the sequence order is injected. There are layer normalization, residual connections, and each position in the feed-forward neural network application. The transformer uses the Encoder-Decoder structure. The encoder layer consists of a multi-head self-attention layer, a position-wise feed-forward network, Residual connections and layer normalization. The decoder layer is similar to the encoder layer with an additional multi-head attention sub-layer. This architecture enabled parallel processing of sequential data while capturing long-range dependencies more effectively than RNNs. The success of transformers in NLP was further solidified by models like BERT (which introduced bidirectional pretraining) and GPT (which demonstrated the power of autoregressive language modeling). These advances raised an important question: Could transformers be adapted to computer vision tasks, where convolutional networks had long been dominant? Yet the Transformer shows limitations that should be considered for any future work. Positional encoding, while effective, may not be the best for modeling long sequences or capturing the inherently hierarchical structures of language. The quadratic self-attention complexity with respect to length remains a problem for very long sequences, although parallelizability of the computation somewhat alleviates this.

Attempts to Integrate Attention into Computer Vision

The Self-Attention Generative Adversarial Network (SAGAN) represents a significant advancement in the field of generative modeling. Published by Zhang et al., this work addresses a fundamental limitation of traditional GAN architectures by introducing self-attention mechanisms that allow the model to capture long-range dependencies in images. The self-attention module computes attention maps that decide where to focus in generating or in evaluating certain parts of an image for their own good fulfilment as well as for coherence. Subsequently, this model learns a global context, unlike its parametrically behaving predecessors. Hence, while generating an image of a dog, it can maintain coherence between the head and the tail even when far apart from each other. The attention-based technical functions in SAGAN thus became not only rewards. The authors have explained deep learning by introducing spectral normalization into both generator and discriminator networks. They further used a unique framework-the use of different update rules for training the discriminator and generator with different learning rates to the two(2) time-scale update rules)-much enhancing training dynamics. They also included a hinge loss version of adversarial loss that did best in the case of high-resolution image generation. Experiments proved SAGAN to be the best performer on the previous leading GANs, mainly on complex scenes with many objects involved. Thus improvements on the model by greatly increasing the Inception Score and reducing Fréchet Inception Distance (FID) were noted-majority of the metrics used for image quality and diversity evaluation for the generated images. Visual confirmation of inspecting extended to outputs also confirmed better handling of geometric constraints and long-range dependencies.

Swin Transformer

Following ViT, researchers proposed improvements to address its limitations: Hierarchical Transformers (Swin Transformer) Liu et al. proposed a new vision transformer, which is named Swin Transformer. Swin is an abbreviation for Shifted Windows. The purpose of this new model is to tackle the challenges that come with using transformers in computer vision. Challenges include that token size is fixed in transformers, which doesn't suit computer vision tasks, and also that images have a higher resolution of pixels than words in text. This approach maintains linear computational complexity relative to image size. The biggest advantage of Swin Transformers is in how they integrate the local feature extraction of CNNs and the global context understanding of Vision Transformers, avoiding the weaknesses of both. Swin Transformers use self-attention mechanisms to understand both local and global relationships. Another strength in Swin transformers is that they require lower computational resources compared to vision transformers. The weaknesses in Swin transformers include Requiring huge computational resources for training, especially for such large datasets. Compared to some of the best CNN-based architectures, it may be poor in understanding very fine-grained local features. The large number of parameters may lead to potential overfitting on small datasets.

MLP-Mixer

In the absence of convolutions and self-attention, Tolstikhin et al. introduced another model used in image classification tasks entitled MLP-Mixer, which is based on multi-layer perceptrons (MLPs). The MLP-Mixer architecture operates on sequences of non-overlapping image patches, similar to Vision Transformers; however, it is carried out in a substantially simplified pipeline. Each input image goes through an operation of being split into several fixed, non-overlapping patches of equal size, which will then undergo flattening followed by a linear projection into some desired hidden dimension. The biggest strength in MLP-Mixer architecture lies in its simplicity. It replaces both attention mechanisms and convolutional operations with simple MLP operations that mix information across either spatial locations or feature dimensions. Another advantage is that MLP-Mixer provides a more computationally efficient alternative to Transformer-based vision models while maintaining competitive accuracy. The visualization of the accuracy-compute frontier confirms that Mixer fits the bill in terms of trade-off between accuracy and compute. The main disadvantage of MLP-Mixer architecture is that while they deliver a good performance, it is not as competitive as the vision transformer performance which reaches better accuracies over different datasets.

Problem Definition

The greatest limitation in vision transformers is the dependence of the model on large-scale pre-training datasets. In the original paper, the focus was on the performance of the model when pretrained on large datasets like JFT-300M, which contains millions of images. This requirement for massive datasets has created a significant barrier due to limited computational resources. However, in this implementation, the focus is on the usage of Vision transformers on small datasets without pre-training on large datasets and without the need for high computational resources. This requires adjusting the hyperparameters in the implementation and modifying the vision transformer components to achieve the best performance, which should be competitive to Convolutional Neural Networks. This project aims to implement a vision transformer by building all its components from scratch, including the patch embedding, self-attention mechanisms, and multilayer perceptron components. The implementation will mostly follow the architecture described in the paper ”An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale” while providing a modular design for the building blocks and model hyperparameter adjustments.

Methodology

The vision transformer consists of three main blocks as shown in Figure 1. The first component is the patch embedding layer, where the image is flattened into linear patches. This block also contains two things, which are position embeddings to maintain spatial information and a class token. The position embeddings are critical for retaining the spatial relationships between patches since transformers do not have any kind of spatial awareness. These embeddings are learned during training and added to the patch embeddings to encode spatial information. The class token functions as an aggregation of sorts for classification tasks. This special token is prepended to the sequence of patch embeddings and serves as the representation of the entire image in the final layer. This block is demonstrated by equation 1 in the paper.

Equation 1 (Patch Embedding):

$$\mathbf{z}0 = \left[\mathrm{\mathbf{x}class}; \mathbf{x}p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}; \ldots; \mathbf{x}_p^N\mathbf{E}\right] + \mathbf{E}\mathrm{pos} \tag{1}$$

The second block is the transformer encoder layer, made up of layers of MLP blocks and multi-head attention blocks, alternating. These two create a feature extraction mechanism that captures both local and global dependencies in the image. The multi-head attention allows the model to focus on different parts of the input simultaneously, creating multiple representation subspaces. Before each alternating block is added a normalization layer, which helps to stabilize training and improve convergence properties. Therefore, the distributions of features are made stable across the entire deep network. This residual connection design helps mitigate the vanishing gradient problem in deep networks. The second among these components is multi-head attention block, shown in equation 2, while MLP block is shown in article equation 3.

Equation 2 (Multi-Head Self-Attention):

$$\mathbf{z}'l = \mathrm{MSA}(\mathrm{LN}(\mathbf{z}l-1)) + \mathbf{z}_{l-1}, \text{ for } l = 1 \ldots L \tag{2}$$

Equation 3 (Feed-Forward Network):

$$\mathbf{z}_l = \mathrm{MLP}(\mathrm{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l, \text{ for } l = 1 \ldots L \tag{3}$$

The last of these things is MLP Head, which maps the learned features of an input to the output class predictions, which will be the predicted number inside the image. This classifier takes the output of the final transformer encoder layer and projects it to the number of classes in the dataset. This last layer of transformation takes that high-dimensional representation of features, reducing it further to class logits. It is only the first token (token 0) that is passed to the

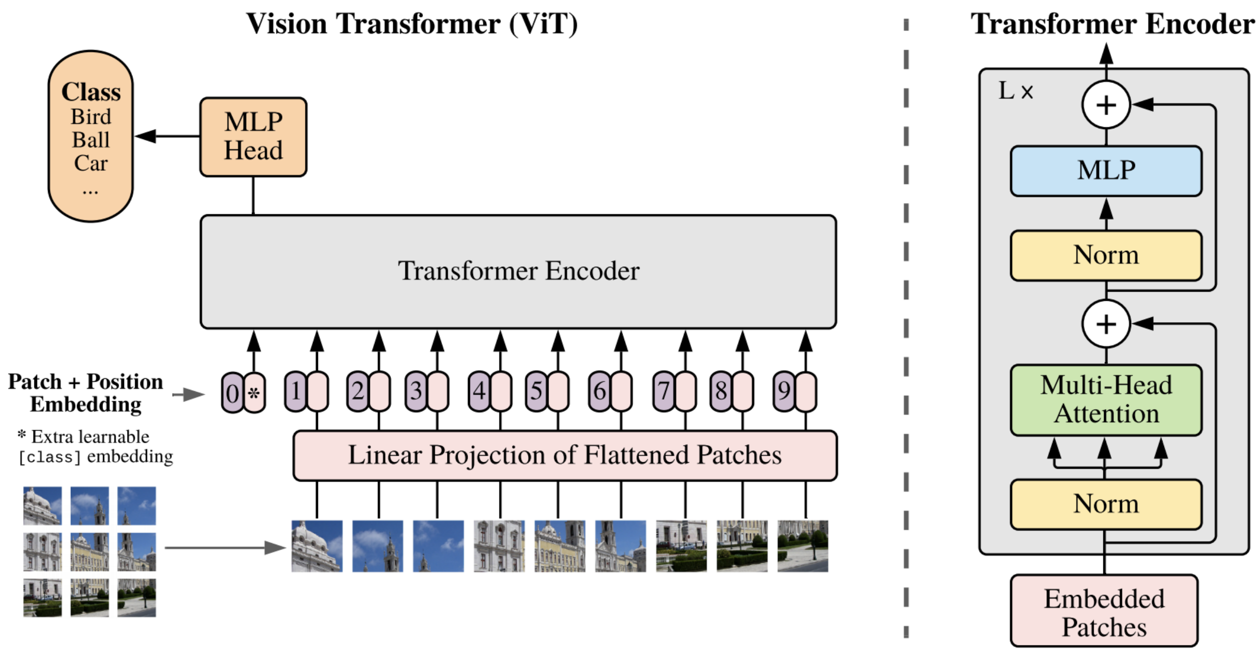

Figure 1. Vision Transformer Model overview

MLP head, as specified in equation 4. This design choice is inspired by BERT's [CLS] token approach. The token continues to act as the representation of the image, similar to the class token of BERT since it accumulates information from every image patch with the attention mechanism. Through self-attention, this token attends to all image patches, effectively aggregating the global context. Combining all these blocks in one class will result in the final model of the vision transformer. The forward method will be simple by making use of only these three blocks in the same order.

Equation 4 (MLP Classification Head):

$$\mathbf{y} = \mathrm{LN}(\mathbf{z}_L^0) \tag{4}$$

The image size is 28 x 28 pixels in the MNIST dataset, and for CIFAR-10, the image size is 32 x 32. So, instead of dividing the image into patches of 16 x 16 pixels as in the original paper, it is divided into patches of 4 x 4 pixels. In addition to one extra learnable class embedding. So, the total number of patches would be 50 patches for MNIST dataset and 65 patches for CIFAR-10. These patches are linearly projected into an embedding space with dimension 256. The transformer encoder contains 8 attention heads in its multi-head self-attention (MSA) mechanism. Regarding the number of Transformer Encoder Layers, the vision transformer consists of 4 Transformer encoder layers. The activation function used in MLP is GELU.

Experimental Results

The implementation of the Vision Transformer architecture described in the Methodology section demonstrated effective learning capabilities when applied to small datasets. All experiments were conducted using identical hyperparameters across both MNIST and CIFAR-10 datasets. The model was trained on an NVIDIA T4 GPU provided by Google Colab and using PyTorch framework, with the following fixed configurations Our training configuration balances efficiency and model optimization. The training process shows stable convergence across 60 epochs, with batch size 256, and using Adam optimization. The Vision Transformer was trained with an initial learning rate of 1e-3, gradually decreasing to 1e-5 through cosine annealing scheduling. A weight decay of 1e-4 was implemented for regularization and standard Adam optimizer parameters were used (0.9, 0.999). Gradient clipping of 1.0 was used to keep training stable, especially in the case of the transformer architecture, which can be vulnerable to exploding gradients. This setting gave a sufficient number of iterations for convergence while ensuring computational efficiency and preventing overfitting.

Datasets

The datasets used are MNIST and CIFAR-10. MNIST contains a training set of 60,000 examples, and a test set of 10,000 examples of grayscale handwritten digits. CIFAR-10 consists of 50000 training color images and 10000 test color images. The total number of classes is 10 for both datasets, as MNIST contains numbers from 0 to 9 and CIFAR-10 also contains 10 classes of color images.

Results on MNIST dataset

The model demonstrates a great performance on MNIST dataset with an accuracy of 99.5% after 60 epochs of training.

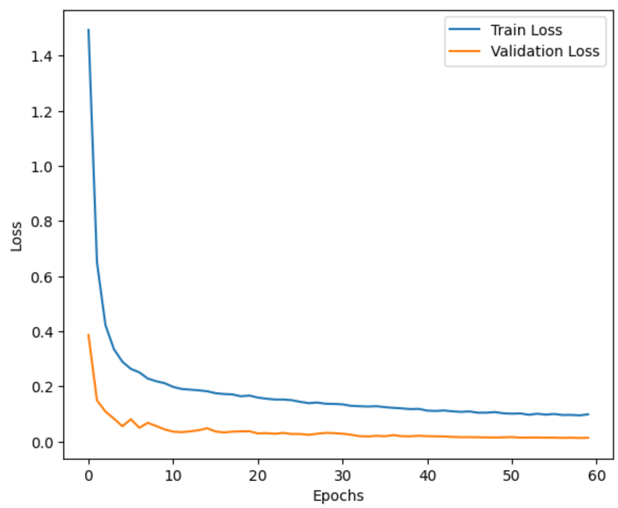

Figure 2. Figure 2 caption

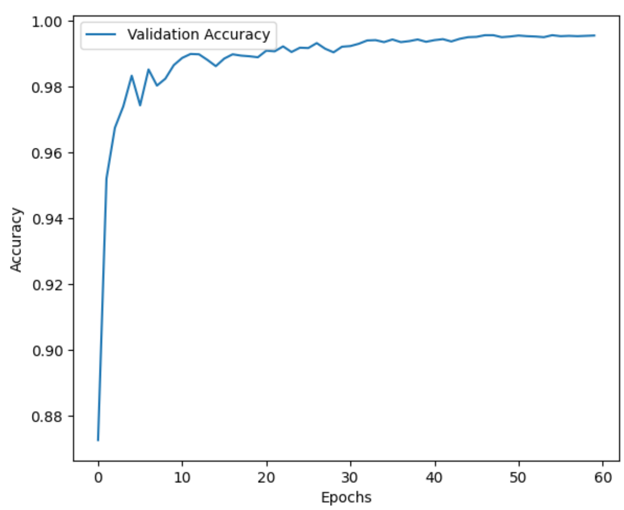

Figure 3. Figure 2 caption

The neural network architecture consisted of 3,313,674 parameters trainable with respect to some modeled data which is around 12.64 MB. This model is designed hierarchically with three major sections comprising the embeddings block with 17,920 parameters independent of encoder blocks totalizing a count of 3,159,040 parameters, and an MLP head with a parameter count of 136,714. This shows that the encoder blocks as the most parametertotaling-intensive component, accounting for about 95% of the capacity of this model. The parameters distribution indicates that Linear layers are dominant with 2,238,986 parameters or 67.57%. It is followed by non-dynamically quantizable linear with 263,168 parameters, which represents 7.94%. Only LayerNorm, with only 4,608, or 0.14%, and Conv2d, with only 4,352, or 0.13%, contributed to this minor distribution.

Results on CIFAR-10 dataset

The model's performance on the CIFAR-10 dataset, while not very impressive like MNIST, still achieves a good accuracy of 81.6% after 60 epochs of training.

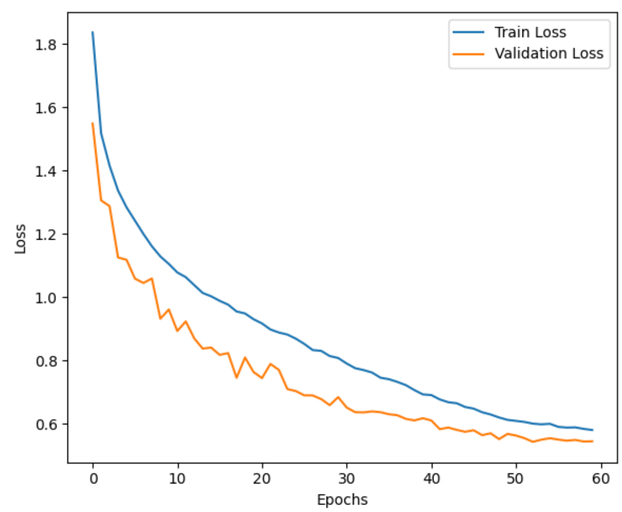

Figure 4. Figure 2 caption

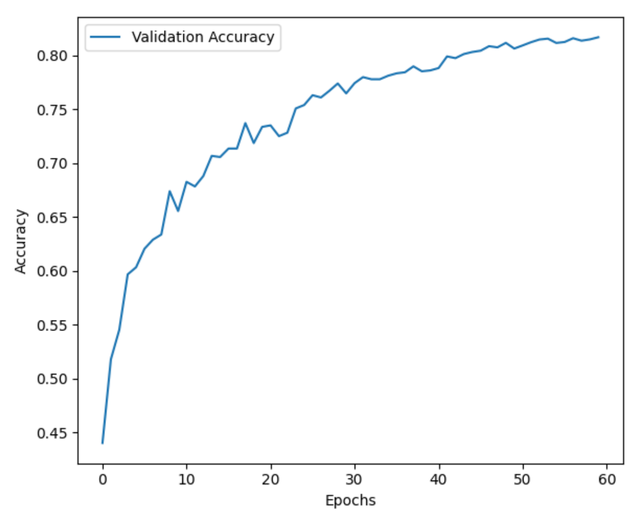

Figure 5. Figure 2 caption

The model architecture contains 3,325,706 trainable parameters with an estimated model size of 12.69 MB. Analysis of the distribution of parameters showed that linear layers formed the bulk of the model parameters (67.32% or 2,238,986 parameters), followed by non-dynamically quantizable linear layers (7.91% or 263,168 parameters). Layers related to convolution and normalization were relatively lightweight, with Conv2d layers accounting for just 0.38% (12,544 parameters) and LayerNorm layers composing only 0.14% (4,608 parameters) of the total number of parameters.

Conclusion and Future Directions

This paper demonstrates that Vision Transformers can achieve strong performance on smaller datasets without requiring large-scale pre-training or extensive computational resources. By modifying the original ViT architecture with smaller patch sizes (4×4 pixels) and optimized hyperparameters, our implementation achieved 99.5% accuracy on MNIST and 81.5% accuracy on CIFAR-10 after training from scratch. These results challenge the common perception that transformers are data-hungry models that require massive datasets to perform well in computer vision tasks. Our experimental results show that the Vision Transformer architecture can be adapted to work effectively with limited computational resources while maintaining competitive performance compared to CNN-based approaches. The model's ability to capture both local features and global dependencies makes it particularly well-suited for image classification tasks, even on relatively simple datasets. However, several limitations and opportunities for future work remain. First, while our model performs well on MNIST and CIFAR-10, further research is needed to evaluate its effectiveness on more complex datasets with higher resolution images and greater diversity. Second, although we reduced the computational requirements compared to the original ViT, there is still room to optimize the architecture further for even greater efficiency without sacrificing accuracy. In conclusion, our work demonstrates that Vision Transformers can be practical for image classification tasks even with limited datasets and computational resources. By carefully adapting the architecture and optimization strategies, we can leverage the strengths of transformer models without their traditional limitations, opening up new possibilities for applying these powerful architectures across a wider range of computer vision applications.

Contribution

This paper extends the core principles introduced by the paper "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale" which focuses on the performance of vision transformers when pre-trained on large datasets like JFT-300M which contains millions of images. However, this paper investigates the application of Vision Transformers (ViT) to image classification tasks on smaller datasets without relying on large-scale pre-training. This requires several adjustments to hyperparameters and architectural components to optimize the model for smaller datasets. One significant factor in this implementation is that it doesn't require extensive computational resources. The model was trained using NVIDIA T4 GPU provided by Google Colab.

LLM Usage Acknowledgement

I acknowledge the utilization of Claude 3.7 Sonnet during this research project. This language model assisted with several aspects of the paper and code development process, while adhering to academic integrity standards and in full compliance with the course's guidelines on acceptable LLMs usage. The model provided support for technical documentation clarification and formatting consistency, and Troubleshooting assistance for implementation challenges. All intellectual contributions, theoretical frameworks, experimental designs, and result interpretations represent my original work. The language model served as a writing assistant tool and to assist in coding and using Python frameworks and debugging.

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